



Technical Service Bulletin: TSB260061

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Revised Air Intake Manifold bracket for QSK23 Air Intake Manifold Heater Coil Fatigue Malfunction

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK23 CM500

Issue

Symptom:

- Low engine power
- White smoke
- Coolant in the lubricating oil

Root Cause:

- The air intake manifold heater coil is malfunctioning because of higher vibration excitation.

Verification/Confirmation

Disassemble the air intake manifold heater from air intake manifold and verify the air intake manifold heater condition.

Resolution

Improved air intake manifold bracket (1), Part Number 6594782 with increased 12 mm thickness along with rib design.

Introduced new air intake manifold bracket (2), Part Number 6594778 with rib design.

Introduced air intake manifold bracket (3), Part Number 6594792 includes mounting location for air intake manifold bracket (2), Part Number 6594778.

Support air intake manifold brackets were added to mitigate air intake manifold heater excitation.

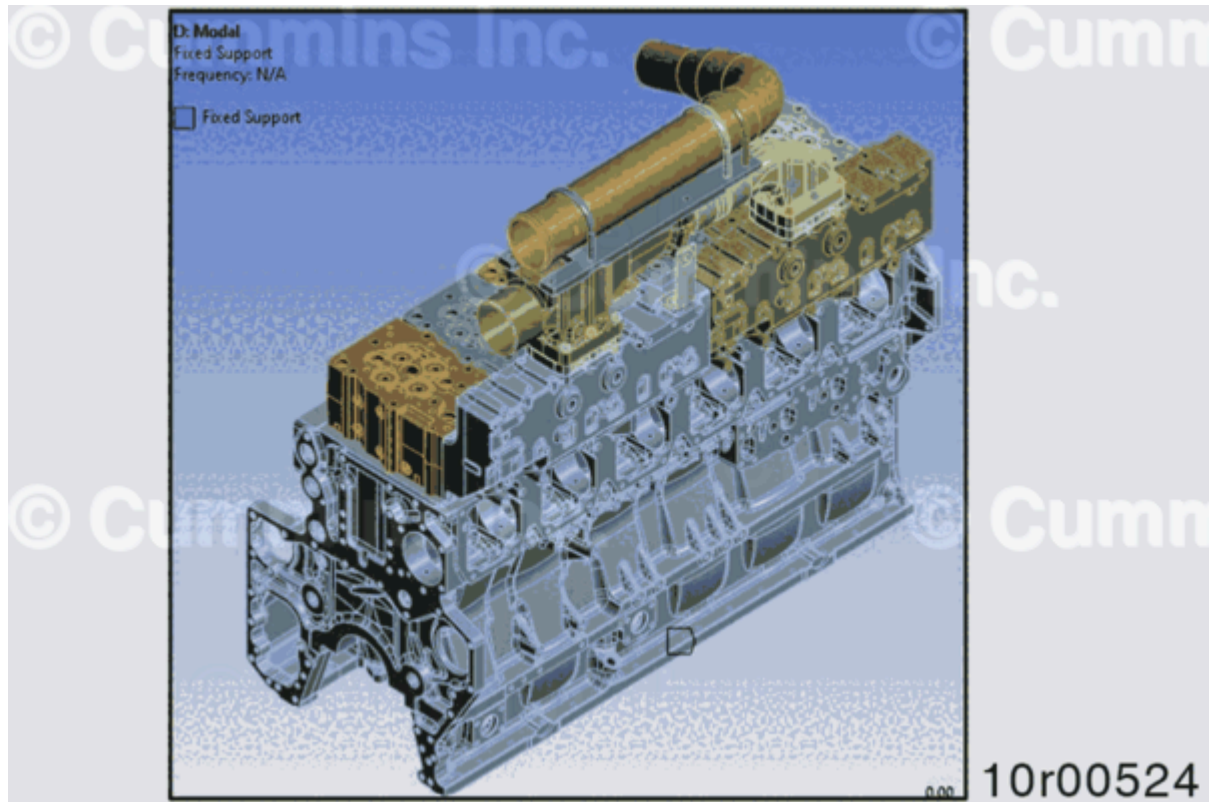


Figure 1, Before air intake manifold bracket installation

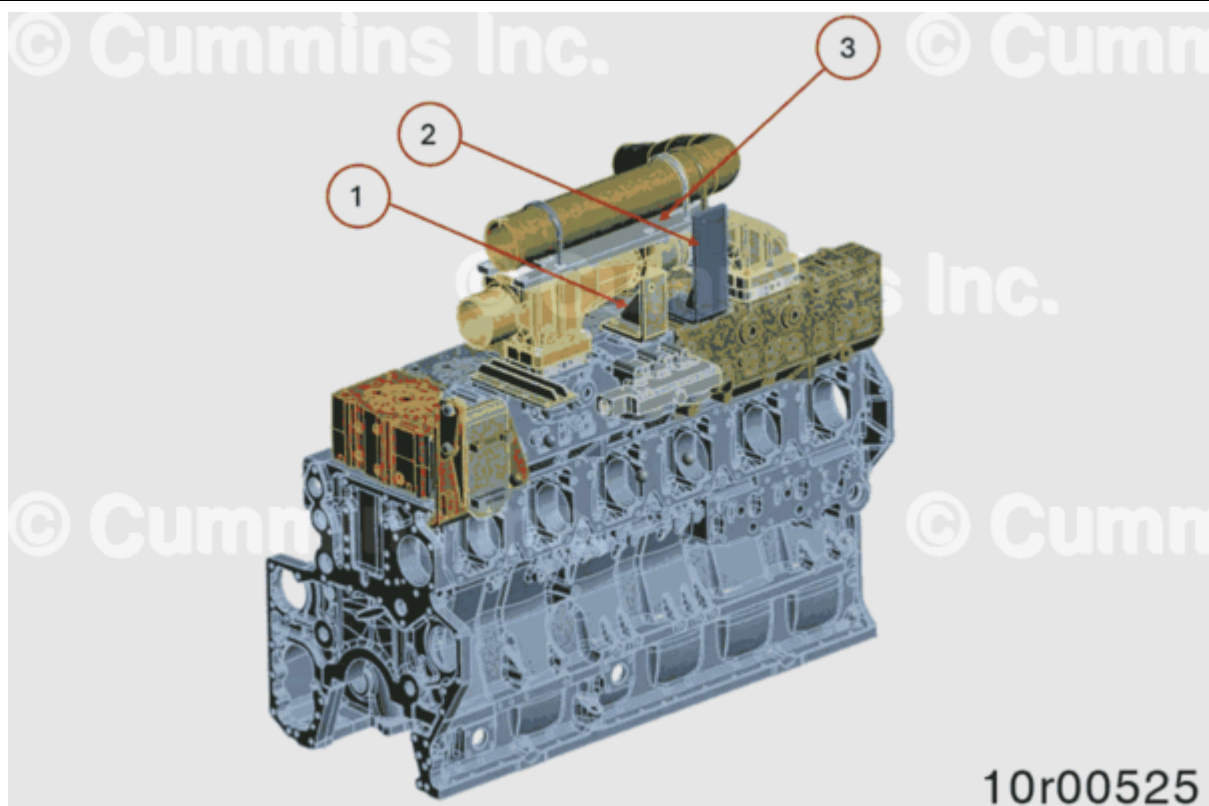


Figure 2, After air intake manifold bracket installation

Description of Change

On existing small air intake manifold bracket (1), Part Number 6594782 thickness increased by 12 mm and addition of gusset. Air intake manifold bracket (2), Part Number 6594778, is newly introduced.

Horizontal plate air intake manifold bracket (3), Part Number, 6594792 modifications for air intake manifold mounting bracket (2).

Reason for Change

Observed air intake manifold heater coil breakage. Damage was then caused to power cylinder components and exhaust path components, leading to low power and white smoke.

Additional air intake manifold brackets added will reduce the vibration excitation of air intake manifold heater coils to prevent malfunctions.



Figure 3, Damaged air intake manifold heater coils

Customer Communication

A design improvement is being implemented by upgrading the air intake manifold bracket to enhance structural integrity and prevent recurrence of the issue.

Service Instructions

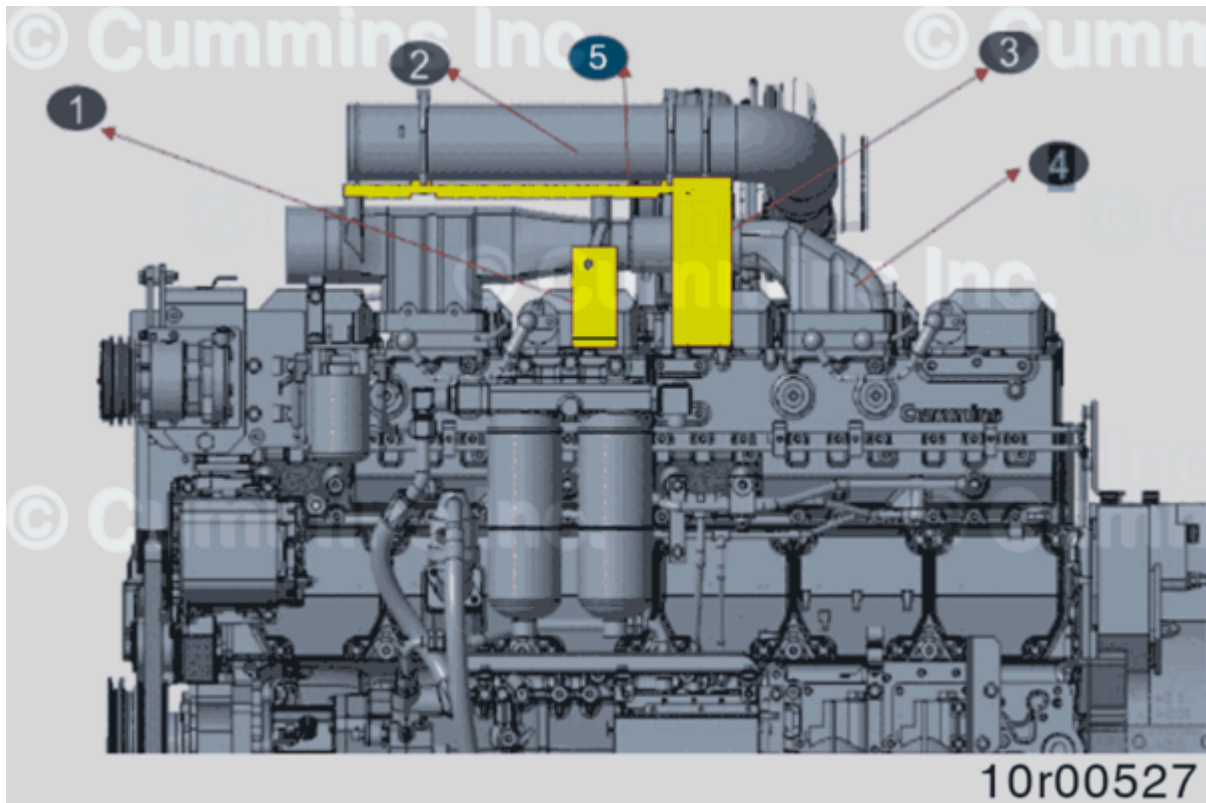


Figure 4, Assembly Details

1. Assemble the rear air intake manifold bracket (3). Torque rear air intake manifold bracket. See Table 1. Assemble the crossover connection (4) as per regular process.
2. Assemble the front air intake manifold bracket (1). Torque front air intake manifold bracket using specification in Table 1. Assemble the air transfer connection tube (2) as per regular process with new plate (5).
3. Torque the rear air intake manifold bracket (3). See Table 1.

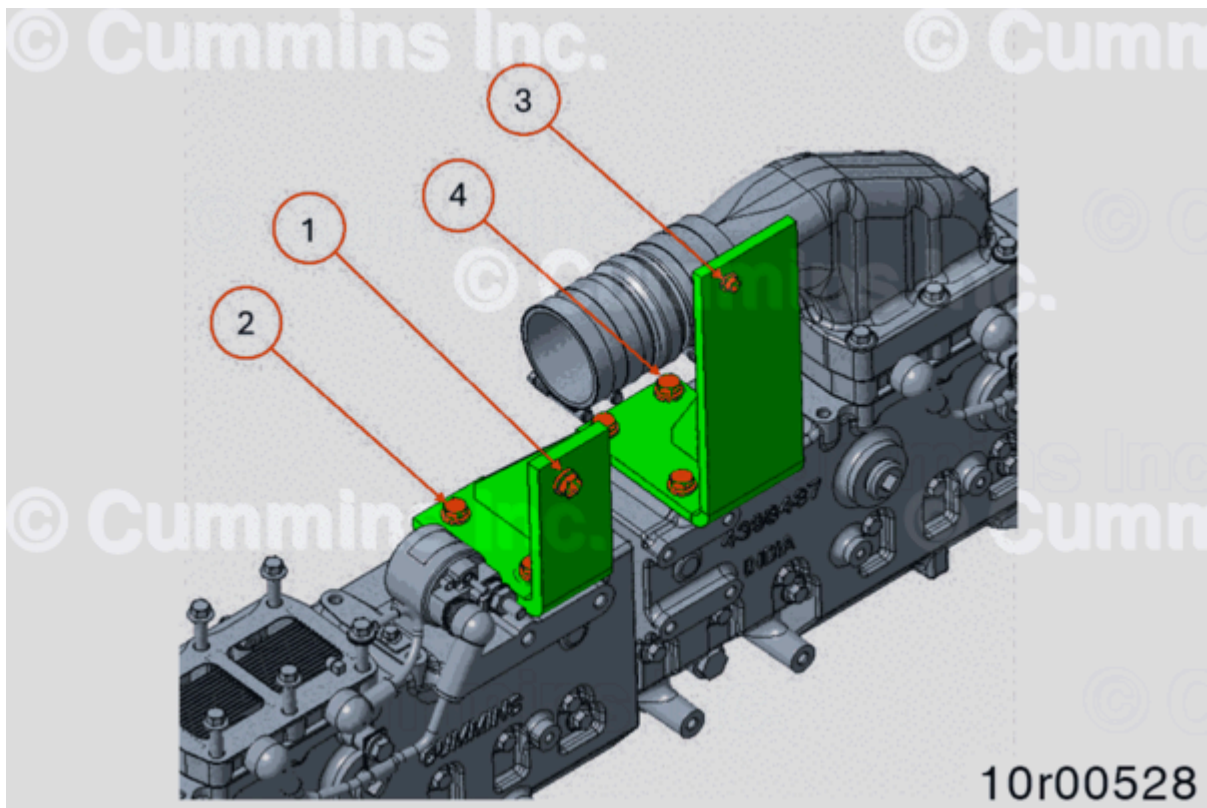


Figure 5, Assembly with Part Numbers Identified.

Table 1, Assembly Specification Details

Item Number	Description	Part Number	Quantity	Specification	Connection	Torque (Nm)	Torque (ft-lb)
1	Captive Washer Cap Screw	4096714	1	M10	Air Intake Manifold Bracket – Crossover	66	49
2	Captive Washer Cap Screw	4096714	2	M10	Air Intake Manifold Bracket – Manifold	66	49
3	Hex Flange Head Cap Screw	3900631	1	M8	Air Intake Manifold Bracket – Crossover	33	24

Table 1, Assembly Specification Details							
Item Number	Description	Part Number	Quantity	Specification	Connection	Torque (Nm)	Torque (ft-lb)
4	Captive Washer Cap Screw	4096714	4	M10	Air Intake Manifold Bracket – Air Intake Manifold	66	49

Service Parts Availability				
Table 2, Service Parts				
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
Connection Bracket	-	-	-	6594778
Connection Bracket	5543958	No	-	6594782
Connection Bracket	4096700	Yes	-	6594792

Part Identification

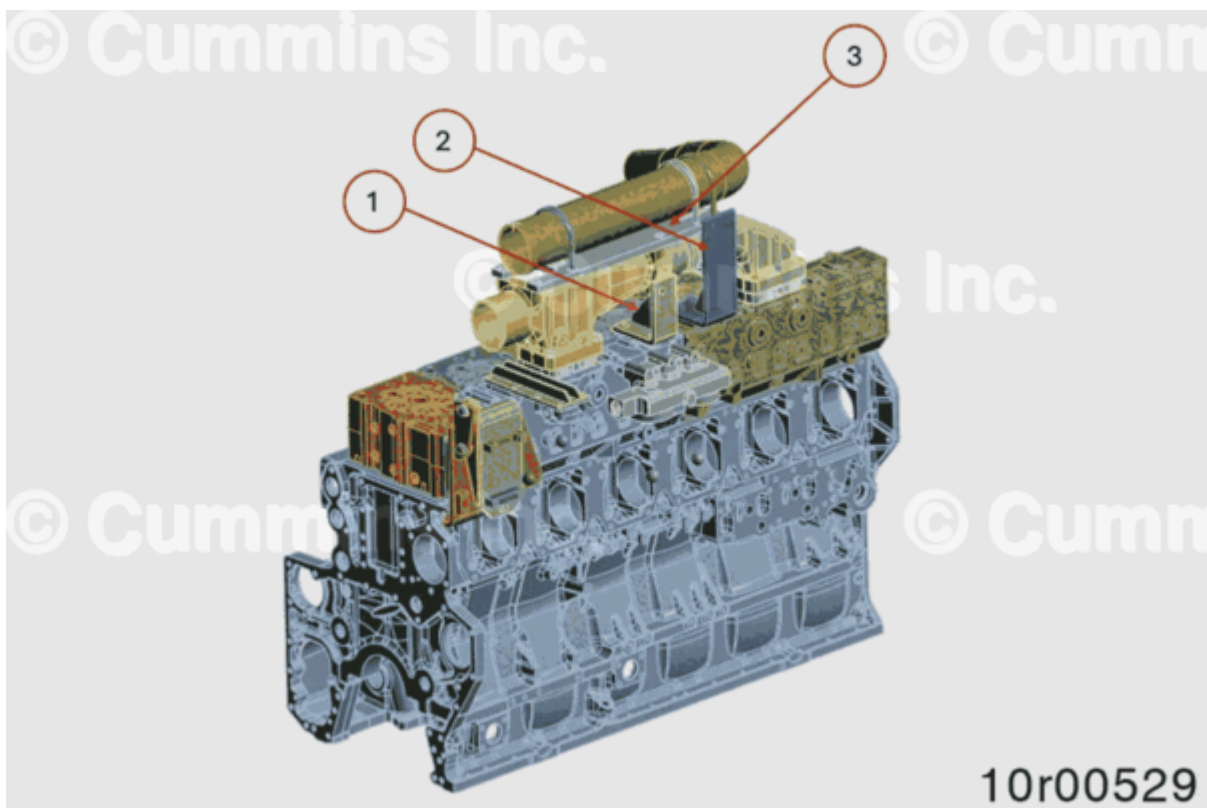


Figure 6, Assembly with Part Number Identified

Table 3, Figure 6: Assembly with Part Number Identified

Item Number	Part Number	Name
1	6594782	Connection Bracket
2	6594778	Connection Bracket
3	6594792	Connection Bracket

Part Inventory Action

Use existing part number inventory before using the new part number.

Production Status

Table 4, Production Information

ESN First	Build Date ¹	Plant
85030297		HHS - CTIL HIGH HORSEPOWER (SEZ) (Phaltan Engine Plant)

¹Engine build date can be found on engine dataplate.

Publications Affected / Associated Publications					
Table 5, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
QSK23 Troubleshooting and Repair Manual	QSK23-CM500	4021375	Air Intake manifold Heater	Install the air crossover connections.	Install

Document History

Date	Details
2026-4-20	Module Created